Computational Identification of Unknown Organic Contaminants and Molecules in Remote Environments via High-Resolution Mass Spectrometry

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1 Goal

This project aims to characterize the chemical footprint of global pollution in remote high-altitude environments by identifying unknown organic contaminants in snowmelt and river water using high-resolution mass spectrometry (HRMS). I hypothesize that long-range atmospheric transport leads to the accumulation of diverse and previously uncharacterized organic molecules in these regions. By combining non-targeted analysis with molecular networking, this study will reveal both the identity and distribution patterns of these contaminants.

2 Specific Aims

2.1 Aim 1: Identify known and suspected contaminants using database-driven screening

- **What (Aim):** Identify known and suspected organic contaminants in high-altitude water samples through HRMS-based suspect screening against curated chemical databases.
- **How (Approach):** Mass spectral features will be matched against databases (e.g., environmental contaminant libraries) using accurate mass and fragmentation patterns to annotate compounds with varying confidence levels.
- **Impact:** This aim will establish a baseline of identifiable contaminants present in remote environments, linking detected compounds to known anthropogenic sources.

2.2 Aim 2: Discover structurally related and unknown compounds using molecular networking

- **What (Aim):** Characterize unknown and structurally related molecules by organizing HRMS data into molecular networks.
- **How (Approach):** MS/MS spectral similarity will be used to construct molecular networks, grouping compounds into clusters that reveal structural relationships and enable annotation of unknowns based on known neighbors.
- **Impact:** This aim will expand detection beyond known compounds, enabling discovery of novel or transformation products and improving chemical annotation coverage.

2.3 Aim 3: Identify patterns in contaminant distribution using unsupervised machine learning

- **What (Aim):** Determine patterns and relationships in contaminant profiles across samples using unsupervised machine learning techniques.
- **How (Approach):** Techniques such as hierarchical clustering and principal component analysis (PCA) will be applied to HRMS data to group samples and identify trends based on chemical composition.
- **Impact:** This aim will reveal underlying structure in the data, helping to infer sources, transport pathways, and environmental factors influencing contaminant distribution.

References

[1] Jules M. Blais, Michael R. Rosen, and John P. Smol, editors. *Environmental Contaminants: Using Natural Archives to Track Sources and Long-Term Trends of Pollution*. Springer, Dordrecht, 2015.

[2] Jordi. Catalan, editor. *Tracking Long-Range Atmospheric Transport of Trace Metals, Polycyclic Aromatic Hydrocarbons, and Organohalogen CompoundsUsing Lake Sediments of Mountain Regions*. Springer, Dordrecht, 2015.

[3] Alex Chao, Jeffrey M. Minucci, Troy M. Ferland, E. Tyler Carr, Greg Janesch, Safia Rizwan, Heather D. Whitehead, Tommy Cathey, Shirley Pu, Laura D. Brunelle, Angela L. Batt, Jon R. Sobus, and Antony J. Williams. Prioritizing chemical candidates from non-targeted analysis using metadata, spectral similarity, and hazard scoring within INTERPRET NTA. *Analytical Chemistry*, 97(29):15904–15912, 2025.

[4] Gillian L. Daly and Frank Wania. Organic contaminants in mountains. *Environmental Science & Technology*, 39(2):385–398, 2005.

[5] Madeleine Ernst, Kyo Bin Kang, Andrés Mauricio Caraballo-Rodríguez, Louis-Felix Nothias, Joe Wandy, Christopher Chen, Mingxun Wang, et al. MolNetEnhancer: Enhanced molecular networks by integrating metabolome mining and annotation tools. *Metabolites*, 9(7):144, 2019.

[6] Wenkai Fan. Suspect screening and semi-quantification of organic contaminants in snow and water samples on the qinghai–tibet plateau and high mountain regions. Undergraduate honors thesis, The College of Wooster, Wooster, OH, 2025. Unpublished thesis.

[7] Peter K.T. Newton S.R. et al. Fisher, C.M. Approaches for assessing performance of high-resolution mass spectrometry–based non-targeted analysis methods. *Anal Bioanal Chem*, 414:6455–6471, 2022.

[8] Rebekah Gray, Eve Painter, Jameson W. Sprankle, Alex D. Crawford, Jillian J. Morrison, Marian Frazier, and Jennifer A. Faust. Suspect screening for pesticides in rain and snow using liquid chromatography high-resolution mass spectrometry. *Atmospheric Environment*, 291:119389, 2022.

[9] B. M. J. Herbert, S. Villa, and C. J. Halsall. Chemical interactions with snow: Understanding the behavior and fate of semi-volatile organic compounds in snow. *Ecotoxicology and Environmental Safety*, 63(1):3–16, 2006.

[10] Tobias Kind, Hiroshi Tsugawa, Tomas Cajka, Yan Ma, Zijuan Lai, Sajjan S. Mehta, Gert Wohlgemuth, Dinesh Kumar Barupal, Megan R. Showalter, Masanori Arita, and Oliver Fiehn. Identification of small molecules using accurate mass MS/MS search. *Mass Spectrometry Reviews*, 37(4):513–532, 2018.

[11] Aliz Kiss and Elke Fries. Occurrence of benzotriazoles in the rivers main, hengstbach, and hegbach (germany). *Environmental Science and Pollution Research*, 16(6):702–710, 2009.

[12] K. E. Manz, A. Feerick, J. M. Braun, Y. L. Feng, A. Hall, J. Koelmel, C. Manzano, S. R. Newton, K. D. Pennell, B. J. Place, K. J. Godri Pollitt, C. Prasse, and J. A. Young. Non-targeted analysis (NTA) and suspect screening analysis (SSA): A review of examining the chemical exposome. *Journal of Exposure Science & Environmental Epidemiology*, 33(4):524–536, 2023.

[13] Stephanie Nino-Suastegui, Eve Painter, Jameson W. Sprankle, Jillian J. Morrison, Jennifer A. Faust, and Rebekah Gray. Non-targeted analysis and suspect screening of organic contaminants in temperate snowfall using liquid chromatography high-resolution mass spectrometry. *Environmental Research*, 266:120494, 2025.

[14] Florent Olivon, Gwendal Grelier, Fanny Roussi, Marc Litaudon, and David Touboul. MZmine 2 data preprocessing to enhance molecular networking reliability. *Analytical Chemistry*, 89(15):7836–7840, 2017.

[15] Robin Schmid, Daniel Petras, Louis-Felix Nothias, Mingxun Wang, Allegra T. Aron, Annika Jagels, Hiroshi Tsugawa, et al. Ion identity molecular networking for mass spectrometry-based metabolomics in the GNPS environment. *Nature Communications*, 12:3832, 2021.

[16] Emma L. Schymanski, Junho Jeon, Rebekka Gulde, Kathrin Fenner, Matthias Ruff, Heinz P. Singer, and Juliane Hollender. Identifying small molecules via high resolution mass spectrometry: Communicating confidence. *Environmental Science & Technology*, 48(4):2097–2098, 2014.

[17] Mingxun Wang, Jeremy J. Carver, Vanessa V. Phelan, Laura M. Sanchez, Neha Garg, et al. Sharing and community curation of mass spectrometry data with global natural products social molecular networking. *Nature Biotechnology*, 34(8):828–837, 2016.

[18] Yue Xu, Gan Zhang, Jun Li, Paromita Chakraborty, Hua Li, and Xiang Liu. Long-range atmospheric transport of persistent organochlorinated compounds to a remote mountain site in southwestern china. *Journal of Environmental Monitoring*, 13:3119–3127, 2011.

[19] Weiting Ye, Jingcheng Li, and Xianfa Cai. MfGNN: Multi-scale feature-attentive graph neural networks for molecular property prediction. *Journal of Computational Chemistry*, 46(3):e70011, 2025.

[20] Weimin Zhu, Yi Zhang, Duancheng Zhao, Jianrong Xu, and Ling Wang. HiGNN: A hierarchical informative graph neural network for molecular property prediction. *Journal of Chemical Information and Modeling*, 63(1):43–55, 2023.